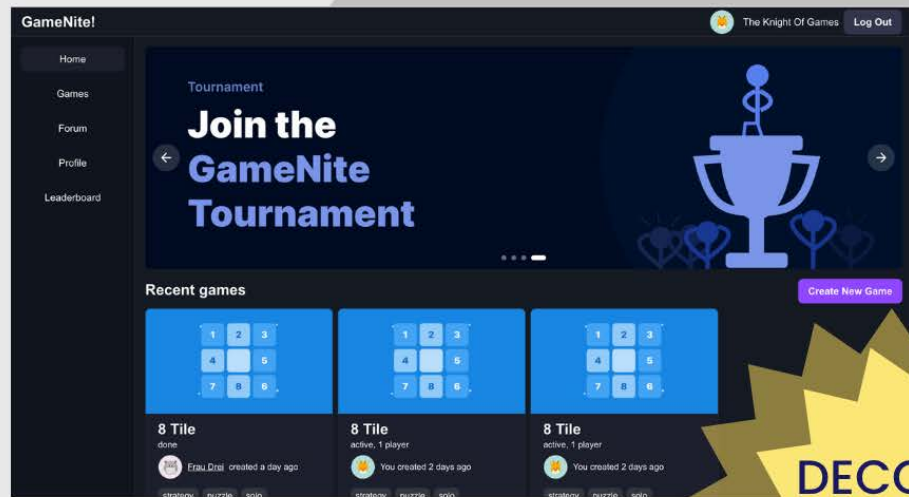
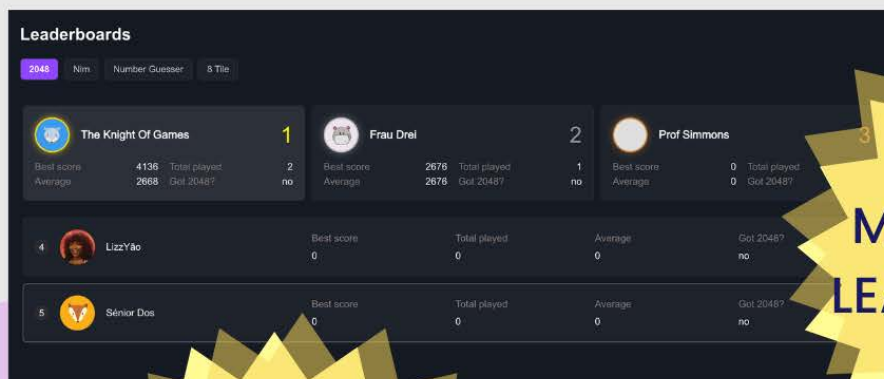


Engaging Gameplay with Personalized Experiences

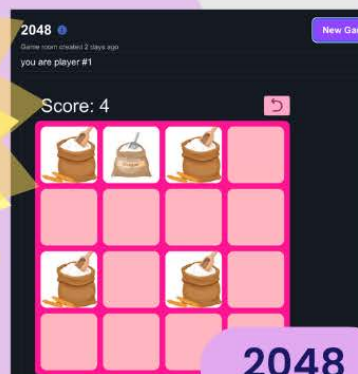


DECORATE
YOUR OWN
PROFILE!

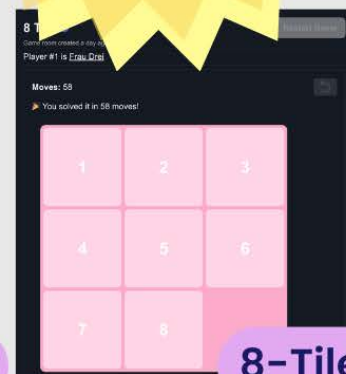


MOVE UP THE
LEADERBOARD!

PLAY NEW
TILE-BASED
GAMES!



2048



8-Tiles

GitHub Repo: <https://github.com/neu-cs4530/spring26-project-spring26-project-group-501>

GameNite Site: <https://spring26-project-spring26-project-group-doop.onrender.com/login>



PLAY POKER WITH FRIENDS!



Watch a Game and talk
to friends in-game

Keep Track of Stats
and Games Played

Play Poker with up
to 8 Friends

REPO LINK

<https://github.com/neu-cs4530/spring26-project-spring26-project-group-502>

PROJECT LINK

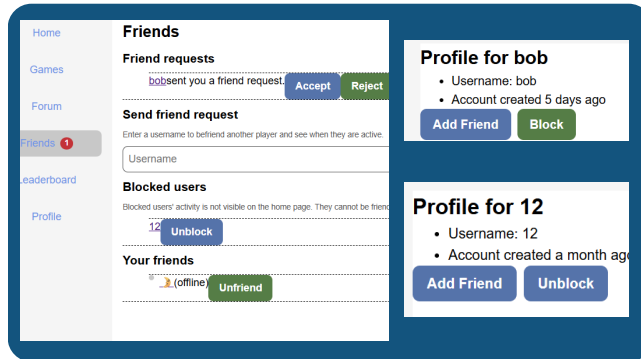
<https://github.com/neu-cs4530/spring26-project-spring26-projecthttps://backend-sp26-cs4530-502.onrender.com/-group-502>

GROUP 503

GameNite - Interactive EFB Edition

Problem Statement

Currently, GameNite's platform lacks the core systems to turn its simple collection of games into a lasting, interactive experience. Players have limited options to build social connections, track competitive growth, or engage in deeper gameplay interactions, and the existing games do not fully showcase the platform's technical potential in UI and responsive design. As a result, GameNite's engagement tends to be session-based, and players have little motivation to return consistently or to invest time in improving their skills.



USER STORY 1

FRIENDSHIP FEATURE

Our friends system will allow users to send and receive friendship requests, as well as to reject or block them. This will allow users to create long-lasting connections and play together

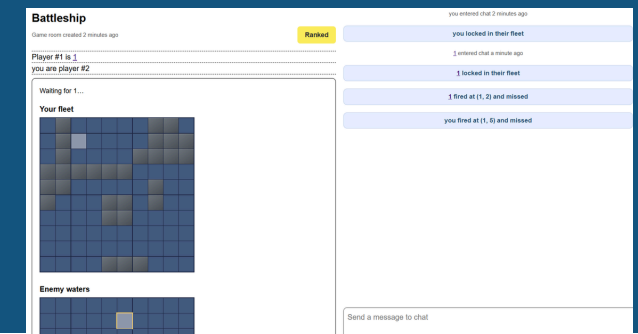
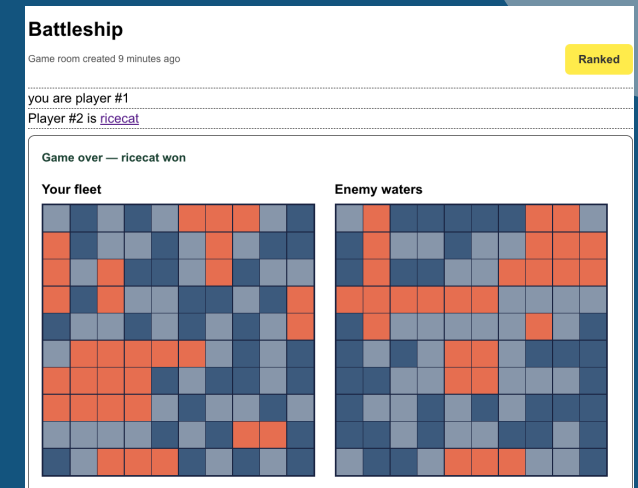
OUR REPO:

OUR DEMO:

USER STORY 3

BATTLESHIP FEATURE

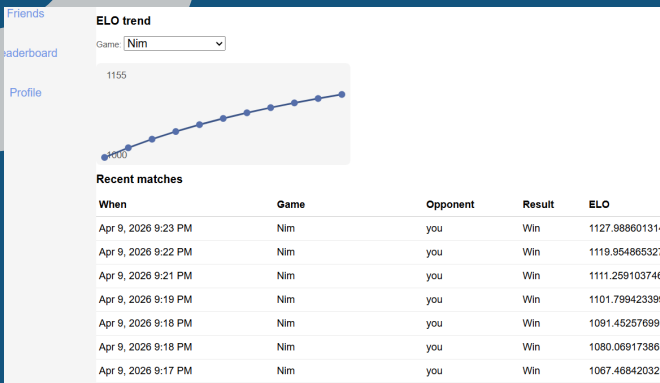
Our Battleship experience will demonstrate the platform's depth by offering gameplay features such as click-and-drag ship placement, real-time feedback on ship placement, a fun background map, and clear indicators of whether an enemy attack was a hit or a miss.



USER STORY 2

ELO RANKING FEATURE

Our ELO ranking system will allow users to view their exact ELO rating and compete against other users of similar ELO ratings. This will provide users with a tangible incentive to improve and grow as a player.



Home	Leaderboard
Games	View: Overall
Forum	Overall
Friends	# Nim ELO Tier
Leaderboard	1 Number Guesser 1161
Profile	2 Blackjack 1068
	3 Yao 1020
	4 S�nior Dos 980
	5 The Knight Of Games 980
	6 you 932
	7 1 839

Your rating
Nim: 844.9989754618285
Number Guesser: 1000
Blackjack: 1000
Battleship: 1000

Making GameNite Social

GROUP 504: NATALIE COSTA, MOAZ GHULMAN, JAE NA WRAY

What we built

A multiplayer gaming platform with a friend system (requests, blocking, mutual friends, DMs, online presence), multiplayer Hangman (up to 6 players, custom/random words by category), and enhanced profiles (avatars, bios, game stats, social links, privacy controls).

Tech Stack

React + TypeScript, Node/Express, Socket.io, MongoDB, Zod shared types. Tested with Vitest + Playwright.

What's next?

Private game rooms, achievement badges, expanded game library, push notifications, and database migration for scalability.

Links

Demo:

<https://spring26-project-spring26-project-group-heh-n.onrender.com>

Repo:

<https://github.com/neu-cs4530/spring26-project-spring26-project-group-504>

US1: Friend System

Friends

[Evan Dini](#) (search) ● Offline Remove

Blocked users

Only you can see this list. These people cannot friend or message you in games until you unblock them.

[Selena Doe](#) (search) ● Offline Unblock

Pending friend requests

[The Knight Of Games](#) (search) ● Online Accept Decline

Outgoing friend requests

1 (0) ● Offline Cancel

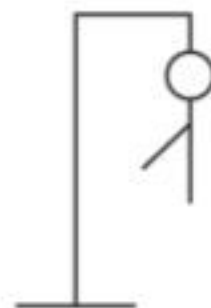
Friends list with blocking, pending requests, and online/offline status

US2: Hangman

Multiplayer Hangman with category-based words and progressive drawing

Hangman

Category: 🐾 Animals



TURTLE

Wrong guesses: 3/6

Correct: E, T, U, R, L

Wrong: A, B, D

🎉 You won! The word was "TURTLE"

US3: Enhanced Profiles

Avatar gallery with selectable profile pictures for personalized identities

Select a profile picture



GROUP 505

Team io

Kaylee Chen, Gavin Sanders, Parth Shah, Grace Li

GameNite evolves from turn-based to real-time multiplayer gaming with Competitive Snake as proof-of-concept!

Global Rankings: SNAKE

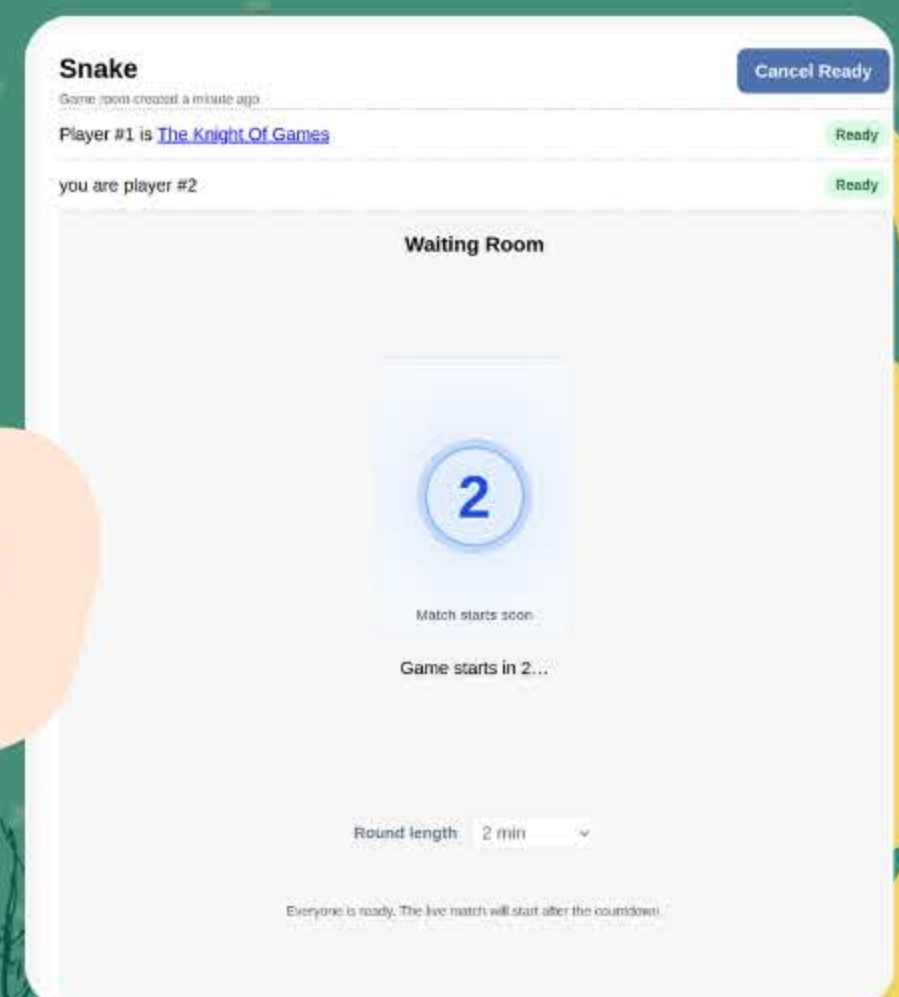
TOP 50 ALL-TIME		
#1	PixelQueen	95
#2	GitchinTheMatrix	95
#3	SpeedRunner2	94
#4	IronClad	91
#5	HealerMain	90
#6	MagiCaster	89
#7	GamerGal	88
#8	ProSnaper	82
#9	RNG_Jesus	77
#10	CaptainCrunch	76
#11	MechaPilot	64
#12	SneakyPete	66
#13	DragonBane	50
#14	ShadowMejia	42
#15	TankBuild	32
#16	LootGoblin	23
#17	NoobSlayer99	15
#18	CosmicDust	12
#19	The Knight Of Games (You)	

Live Scoreboard

Player	Items
The Knight Of Games	0
You	
Yáo	0

**Live + All-Time
Leaderboard**

Waiting Rooms!



Tech Stack

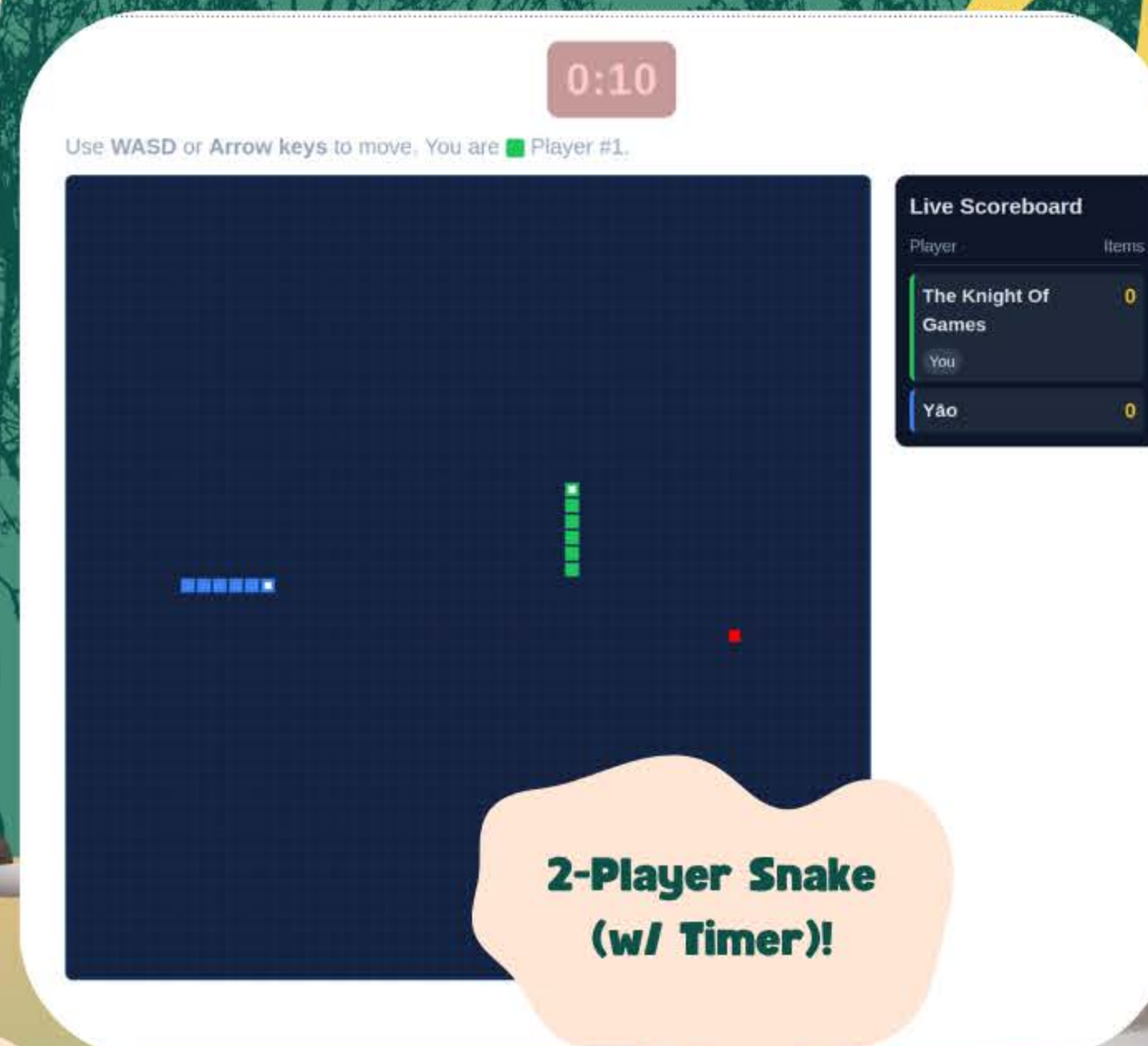
WebSockets, server-side tick rates, grid-based movement, and reusable architecture for future games.

Frontend:

Backend:

What's Next

Additional real-time games, enhanced matchmaking, improved scalability, and modular refactoring.



**2-Player Snake
(w/ Timer)!**

**Checkout out our
PUBLIC DEMO & REPO**

<https://spring26-project-spring26-project-group-lr7c.onrender.com/login>
github.com/neu-cs4530/spring26-project-spring26-project-group-505

GameNite!

AI-Powered Mafia & Competitive Leaderboards

Michael Sica · Emir Durakovic · Min-Hong Shih · Harshul Agarwal

PROJECT FOCUS

This flier highlights our two main additions to GameNite: a fully playable Mafia game with AI-controlled participants, and a leaderboard system that tracks player performance over time.



FEATURE ONE

Mafia Game

A fully functional Mafia party game built from scratch with real-time multiplayer support.

- Lobby creation & role assignment
- Discussion & voting phases with live timers
- Elimination logic & win condition detection
- Real-time chat via WebSockets



FEATURE TWO

AI Agents

8 unique bot personalities powered by **4 LLM providers** (GPT, Gemini, Grok, Claude) which are used fill empty slots so a user always has the ability to play Mafia.

- Contextual, personality-driven chat
- Analytical, provocative, diplomatic & more
- Vote during rounds like real players
- Session memory for tracking suspicion
- Replaceable by humans mid-game



FEATURE THREE

Leaderboard

A global ranking system tracking performance across all games on the platform.

- Points for wins, deductions for losses
- Win rate, games played, per-game stats
- Sortable columns & time-range filters
- Top-3 podium with monthly resets

Rank	Player	Points	Win	Loss	Win Rate	W/L Ratio	W/L Ratio	W/L Ratio
1	AI_001	10	5	5	50%	1	1	1
2	AI_002	8	4	4	50%	1	1	1
3	AI_003	7	3	4	43%	1	1	1
4	AI_004	6	3	3	50%	1	1	1
5	AI_005	5	2	3	40%	1	1	1
6	AI_006	4	2	2	50%	1	1	1
7	AI_007	3	1	2	33%	1	1	1
8	AI_008	2	1	1	50%	1	1	1

Leaderboard Rankings

Player	Role	Status	Points	Win Rate	W/L Ratio
AI_001	AI	Alive	10	50%	1
AI_002	AI	Alive	8	50%	1
AI_003	AI	Alive	7	43%	1
AI_004	AI	Alive	6	50%	1
AI_005	AI	Alive	5	40%	1
AI_006	AI	Alive	4	50%	1
AI_007	AI	Alive	3	33%	1
AI_008	AI	Alive	2	50%	1

Mafia — AI Bots In Action

Player	Role	Status	Points	Win Rate	W/L Ratio
AI_001	AI	Alive	10	50%	1
AI_002	AI	Alive	8	50%	1
AI_003	AI	Alive	7	43%	1
AI_004	AI	Alive	6	50%	1
AI_005	AI	Alive	5	40%	1
AI_006	AI	Alive	4	50%	1
AI_007	AI	Alive	3	33%	1
AI_008	AI	Alive	2	50%	1

End of Game State

WHY WE BUILT THIS

The Problem

Mafia needs a minimum number of players to be fun, but finding 10 people online at once is rare. Our AI agents seamlessly fill empty slots and is able to do all functionality that a human might including chatting, voting, and strategizing—so every game feels complete. The leaderboard also adds a competitive layer that keeps the community coming back by encouraging healthy competition between friends .

TECHNOLOGY STACK

Design & Architecture

A monolithic repository with shared types across client and server.

- **Frontend:** React 19, Vite, React Router, Socket.IO Client
- **Backend:** Express 5, Socket.IO for real-time game state
- **Database:** MongoDB via Keyv for persistent storage
- **AI:** OpenAI SDK with Zod-validated structured outputs; unified adapter across GPT, Gemini, Grok & Claude
- **Testing:** Vitest + Playwright E2E
- **Validation:** Zod schemas in shared package

WHAT'S NEXT

Future Work

We aim to extend AI bots to other GameNite titles (Nim, Guessing) so all the other games can be played solo as well. Furthermore, we aim to add deeper personality systems with cross-session learning so the AI models are able to learn and excel from previous experiences. Lastly, we would like to explore voice-chat integration during discussion phases and expand the leaderboard with achievements and seasonal rewards such as 24-hr point multipliers and in-game powerups.

LIVE DEMO WEBSITE

spring26-project-spring26-project-group-y8vc.onrender.com

SOURCE REPOSITORY

github.com/neu-cs4530/spring26-project-spring26-project-group-506

Murder on Game Nite

Team 507

Sharyq Siddiqi, Kiki Liu
Lucas Balfanz, Declan Javier

<https://spring26-project-group-507.onrender.com/>

<https://github.com/neu-cs4530/spring26-project-group-507>

Design Decisions



Murder on Game Night: To create Murder, we needed to find a way to code a game that updates on a timer, rather than with manual player actions. To accomplish this, we created an invisible 'host' player, driven by the server, which joins as a player and can send periodic game updates as 'moves' through the same pipeline as player moves, allowing it to fit neatly within the existing game framework.

Asymmetric gameplay was the key design challenge. Player actions were separated into their own file which 'queued' them, and determined the outcome all at once at the end of the phase.

Chat: Each player has a dedicated chat room on the backend, and messages include an optional recipient list so the server can route them to specific users. The chat controller also enforces phase-based rules—restricting who can communicate during each stage and blocking dead players from messaging the living.

Notes: Notes are stored in a database keyed by both game ID and user, so each player's notes are private and game-specific. The backend mirrors the existing chat architecture—using the same socket and controller patterns—and the UI lives in the chat panel, toggled via a 'notes' tab to save screen space.

Features

Murder On Game Night: Our primary goal was to build a game similar to Werewolf, where a group of townspeople are picked off by a murderer, and the townspeople need need to figure out who it is. Players are randomly assigned roles, which can have unique actions. Our implementation features timed phases for chatting, voting, and using player actions, as well as our own spritework.

Chat: To make the game more intuitive, we modified the chat to support personal/group messages, which can be sent to individual or multiple players.

Notes: We also added a 'notes' tab, which allows you to write down information and save it for future reference. You have a different set of notes for each game.

What's Next

There is a lot of room to continue our work on this project. The framework exists to expand the game both visually, through adding more sprites for different actions, as well as in the backend, by adding more roles.

The chat and notes could also be expanded, such as adding dedicated tabs for group chats, and game-specific Notes UI.

Making GameNite Competitive

By: Nate Gordon, Henry Bednarz, Jolin Wang, Alejandro Hernandez

Challenge Mode

Challenge Pick/Ban Phase

Players take turns banning games until the randomizer selects one from the remaining pool!

Available Games to Ban

Ban Nim

Ban Number Guesser

Ban Concentration

Ban Liar's Dice

Ban Rock Paper Scissors

Ban Chopsticks

Banned Games

- None yet.

21s remaining

It is your turn to ban a game!

- New way to play which encourages players to be skilled across all game modes!
- Turn timer forces players to think quickly and finish games.

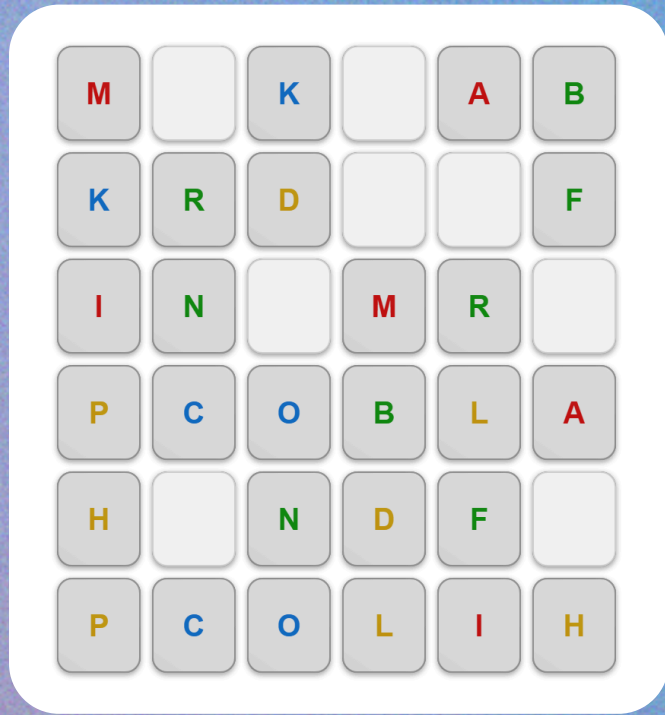
Leaderboard

1 point per opponent beaten.			
Rank		Player	Points
#1	↑ 1	Air Mail (You)	11
#2	↑ 2	Yāo	10
#3	↓ 2	The Knight Of Games	10
#4	↓ 1	Blue Cherry	7
#5		Frau Drei	4

- Site-wide ranking system made to track your performance.
- Rankings based on points, awarded for beating opponents in completed matches.

[Play Here!](#) [Our Repo!](#)

Competitive Games



- Four new fun and competitive games to play!

Liar's Dice

It is currently **your** turn.

Current bid: 3 dice showing 3

Players

- Yāo — 5 dice
- you — 5 dice

Your dice

2 3 3 4 5

Your move

Quantity:

Face:

Bid History

- Yāo bid 1 × 1
- you bid 3 × 1

Why These Features?

- By creating a competitive environment on GameNite we can ensure players will keep coming back!

What's Next?

- Side-wide Tournaments, More engaging UI, and more games!

Design Choices

- 1) Leaderboard: Uses existing win record data, along with snapshot storage for tracking current position and change.
- 2) Challenge Mode: Wraps GameNite's existing game logic with pick/ban, timers, and a centralized game registry.
- 3) Followed existing game architecture.

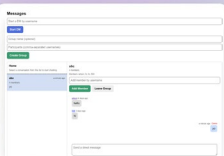
Tech Stack

- TypeScript
- React
- Vite
- Express
- Socket.io
- Zod
- Playwright
- MongoDB

Team 508

GameNite

Play. Chat. Customize.



Chat in Real Time

Direct and group messaging with live typing indicators, read receipts, and instant delivery powered by WebSockets.



Chess, Reinvented

A full-featured chess game with Stockfish AI opponents at configurable difficulty and per-player timers.

Profile

General information

- Username: hi
- Account created 6 days ago

Stats

- Wins: 10
- Losses: 4
- Draws: 0
- Win rate: 71.4%

Badges: 🏆 First Victory, 🎓 Nim Novice, 🎲 Guess Novice, ⚔️ Veteran



Stats & Achievements Track Your Glory

Persistent win/loss stats, 12 achievement badges across global and per-game milestones, with full privacy controls.

Theme customization

Try colors here before saving them to your profile.

Theme mode

Light

Primary color



Secondary color



Tertiary color



Apply preview

Reset theme

Game night profile

Wins: 10

Win rate: 71%

Primary

Secondary

Tertiary



Theme Customization Make It Yours

Light and dark mode toggle plus a color picker with live preview so every user can personalize their experience.



RANKED ROYALE: COMPETITIVE MATCHMAKING & LEAGUE PROGRESSION

Group 510: Dan Nguyen, Jacob Shechter, Mika Tums, Arshia Verma

Northeastern University

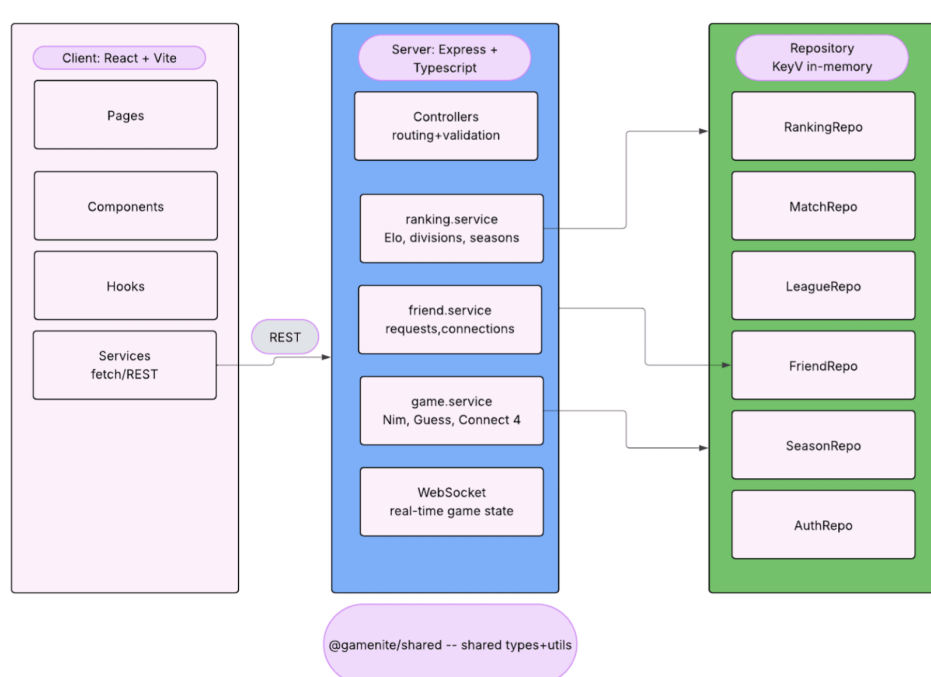


Introduction/Abstract

GameNite is a multiplayer gaming platform that supports casual play across several skill-based games. While the platform provided a solid foundation for social gaming, it lacked any structured competitive system: wins and losses had no lasting meaning, and players had no incentive to return regularly. This project introduces **Ranked Royale**, a competitive ecosystem built on top of GameNite featuring performance-based *Elo matchmaking*, *seasonal league divisions*, *a new game (Connect 4)*, and *a friends system with social play features*. The goal was to transform GameNite from a casual platform into one where competitive players can track progression, climb divisions, and challenge friends, while preserving the existing casual experience for players who prefer it.

System Design/ Methodology

The system was built as a three-layer architecture: a React + Vite client communicating via REST API to an Express server, which delegates to an in-memory KeyV repository layer. The ranking system uses the Elo formula with a flat K-factor of 32, a well-established standard for players that provides meaningful rating movement per match. Players are assigned to one of five divisions: Bronze, Silver, Gold, Platinum, and Diamond based on their Elo rating, with separate ranking records maintained per game per season to prevent cross-game contamination.



To the left here is the architecture diagram. We have the Client -> Server -> Repository.

Matchmaking enforces division-based pairings so players only compete against opponents of similar skill, with an exception for friend challenges which bypass this restriction to preserve social flexibility. The friends system uses WebSocket push notifications rather than REST polling, eliminating the server overload caused by repeated HTTP requests.

Division-Based Matchmaking: Players can only join open ranked matches created by players in their own division. This ensures competitive fairness, for example: a Bronze player cannot be matched against a Diamond player. This also mirrors standard competitive design. Friend challenges, however, bypass this restriction intentionally. For example: if Yao is friends with Knight of Games (KOG), and Yao is in the Gold division, and KOG is in Silver, KOG can't load a match that Yao made in the Ranked Match section. However, if Yao deliberately sends a ranked match game to KOG through Friends, KOG can join and play with Yao. *This feature preserves social play flexibility.*

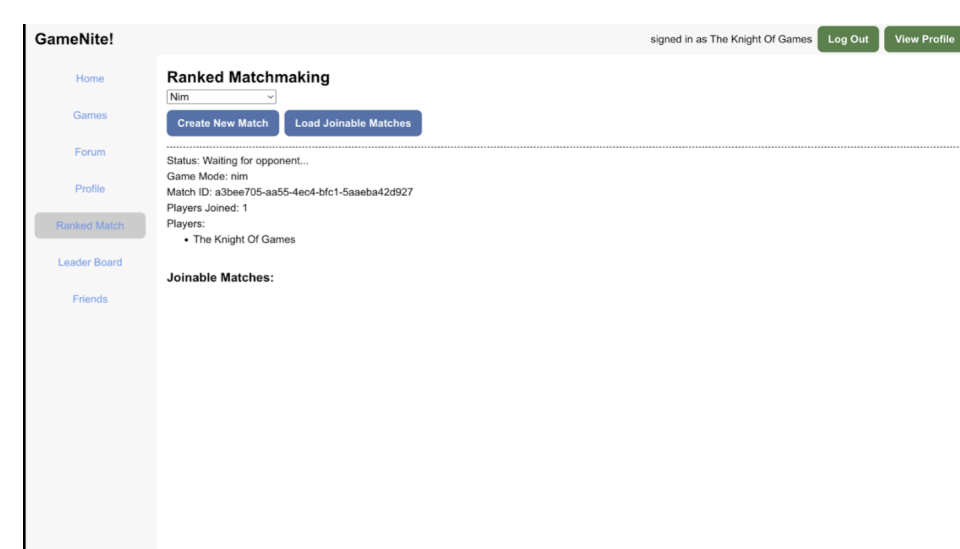


Fig. 1: Creating and joining a Ranked Match.



Collaboration

The team followed an agile sprint-based workflow across four sprints, with two recurring weekly meetings: one internal team sync and one with the course TA to review progress and surface blockers early. All development was managed through GitHub Projects using a Kanban board with issues assigned per sprint, giving the team shared visibility into what was in progress, blocked, or complete at any time.

All code was developed on feature branches and merged via pull requests, with a required review from at least one teammate before merging into main. Code reviews were used not just for catching bugs but for knowledge sharing. Reviewers left inline comments explaining design concerns, suggesting refactors, and flagging edge cases. This kept the entire team aware of changes happening outside their own tasks and prevented integration surprises at the end of each sprint.

Features & Implementation

Three user stories were fully implemented. For ranked matchmaking and league progression, players can create or join ranked matches, track their Elo rating per game, view division standings, and see their full match history with win/loss indicators and Elo changes. For new competitive games, Connect 4 was implemented with full game logic, real-time WebSocket state updates, and both ranked and casual play modes. For the friends system, players can send and accept friend requests, view a friend's competitive profile including ranked stats and match history, and directly challenge friends to ranked matches regardless of division. All features are live at the deployed URL.

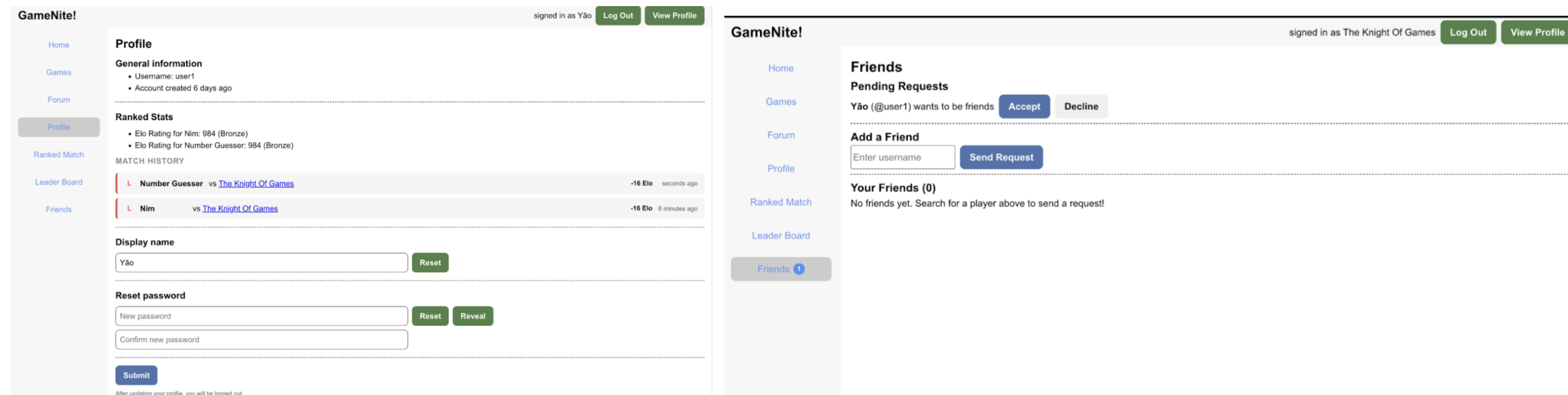


Fig. 3: Profile page showing Elo ratings and match history, ranked matchmaking page, friend profile page showing ranked stats

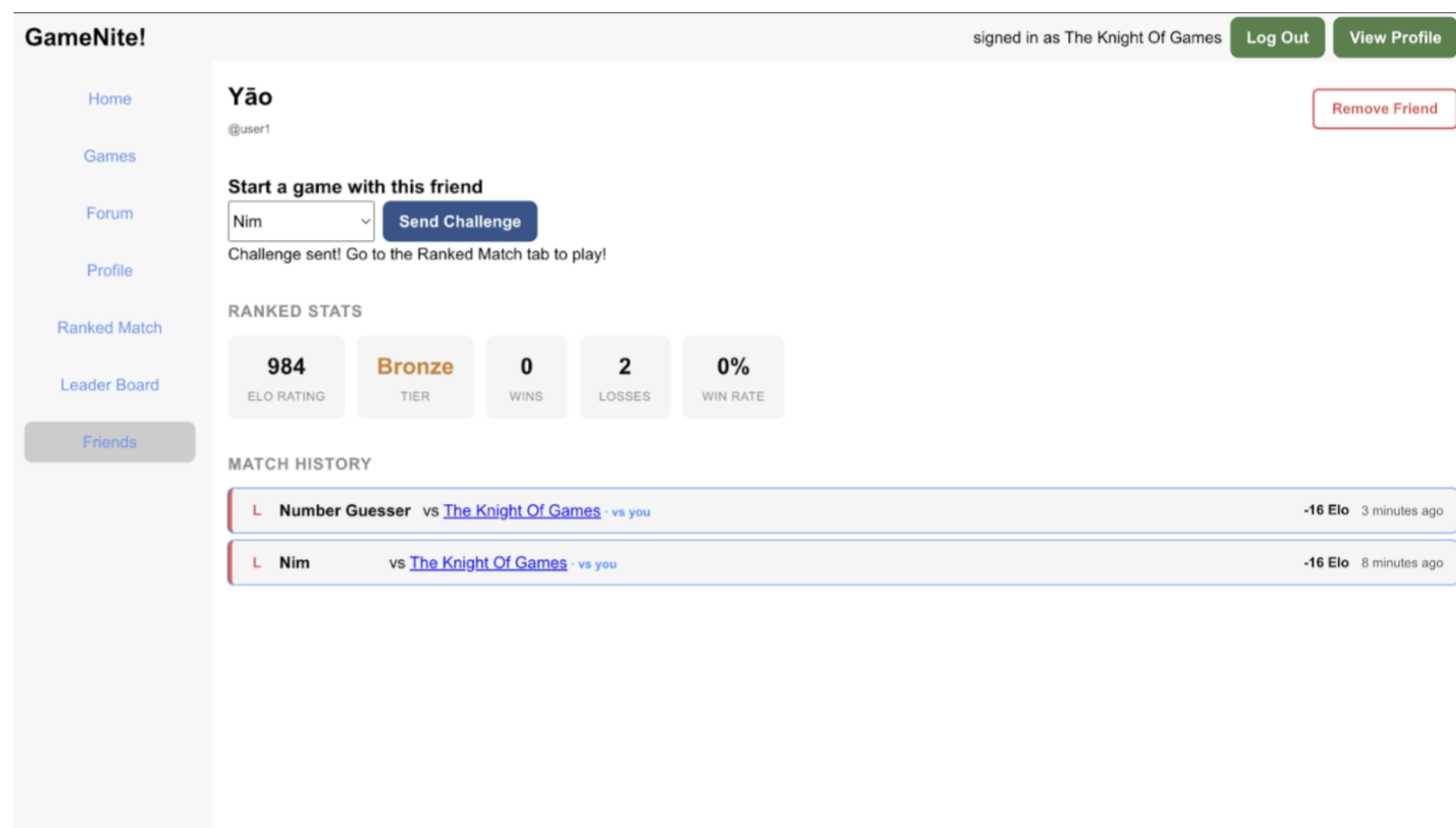


Fig. 4: View Friend Match History + Send Challenge

LINKS TO GITHUB REPO AND HOSTED APP

Hosted Render app: [Here](#)

GitHub repo: [Here](#)

Discussion/Future Work/Web App

The project successfully delivered a competitive layer on top of GameNite with all essential conditions of satisfaction met across three user stories. Several desirable features were descoped due to time constraints, including automatic season rotation, end-of-season division movement, a friends-only leaderboard, and side-by-side stat comparisons. A notable mid-sprint deviation was the refactor of the friends polling system to WebSockets after identifying that repeated REST polling was causing 502 errors on the free-tier Render deployment: a real-world constraint that shaped the final architecture. Future work would focus on activating the existing season infrastructure with a scheduled rotation job, adding rank badges to player profiles, and implementing a bracket-style tournament mode that was originally planned but replaced by the friends system.

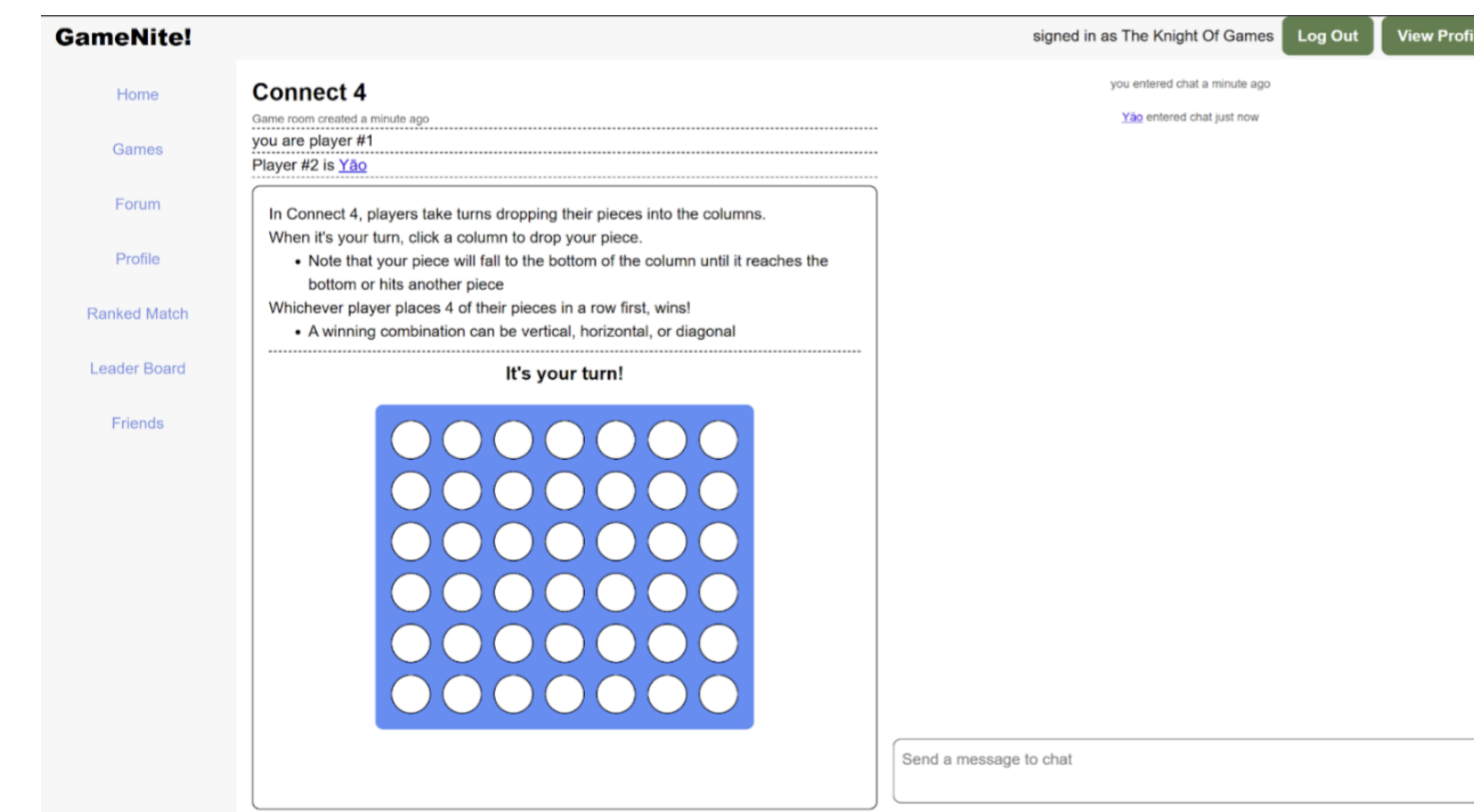
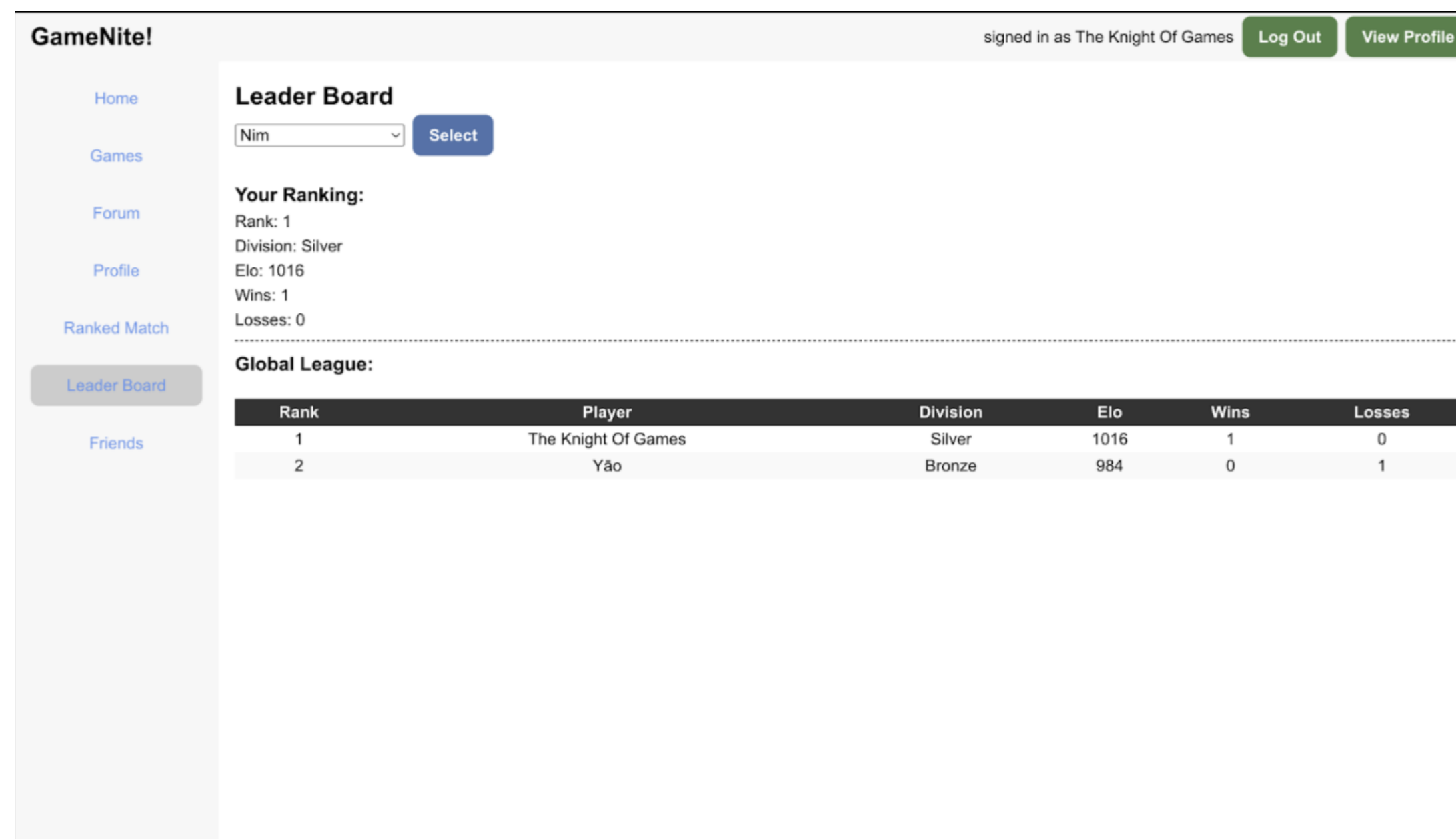


Fig. 5: Connect 4 Frontend & Leaderboard to Select and View Standings



User Stories

User Story 1: Ranked Mode/Matchmaking + League Progression

- As a competitive player, I want to play ranked matches that affect my statistics and rating, so that I can measure my skill progression and compare myself to other players.

User Story 2: New Competitive Games

- As a competitive player, I want access to a variety of skill-based games to compete in, namely Connect Four, so that ranked play remains diverse and engaging

User Story 3: Friends System

- As a social and competitive player, I want to add friends and view their activity, so that I can keep up with their competitive progress, compare rankings, and play games against them.



GameNite transformed from a simple game platform into a social, competitive experience focused on community-building through real-time multiplayer interaction, more expressive communication, and competitive progression.

you entered chat seconds ago

Player seconds ago

@yaos hey man, this is an example of a mention

Player just now

Hello again! This is an example of markdown rendering

WHAT'S NEXT

- Host-configurable rules (custom roles, extra Mafia members) in the lobby when creating a game
- Friend lists, invite-only lobbies, and game history
- Tier badges and player search on the leaderboard

OVERVIEW

Core Theme

All features center on COMMUNITY:

- Mafia: collaborate & bluff
- Chat: GIFs, markdown
- Leaderboards: competition

Tech Stack

- MERN Stack (MongoDB, Express, React, NodeJS)
- Fully written in TypeScript
- RESTful API + WebSockets for communication
- Infra: Render.com (Web Hosting), MongoDB Atlas (Hosted MongoDB), Github Actions (CI)
- Services: Giphy API



<https://www.github.com/neu-cs4530/spring26-project-group-511>



<https://cs4530-s26-511.onrender.com>

Mafia

Game room created a minute ago
Player #1 is [The Knight Of Games](#)
Player #2 is [Yao](#)
you are player #3
Player #4 is [Frau Drei](#)
Player #5 is [Lebron James](#)

Night Phase

Your role: Mafia 30s remaining

The Knight Of Games

Kill

Yao

Kill

Sénior Dos (you)

Frau Drei

Kill

Lebron James

Kill

Mafia Wins

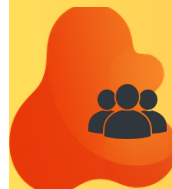
Your role was Nurse.

You lost this round.

Alive: 2 Eliminated: 2 Players: 4

Player #1	Cop	ELIMINATED
Player #2	Nurse	ELIMINATED
Player #3	Mafia	ALIVE
Player #4	Civilian	ALIVE

FEATURES



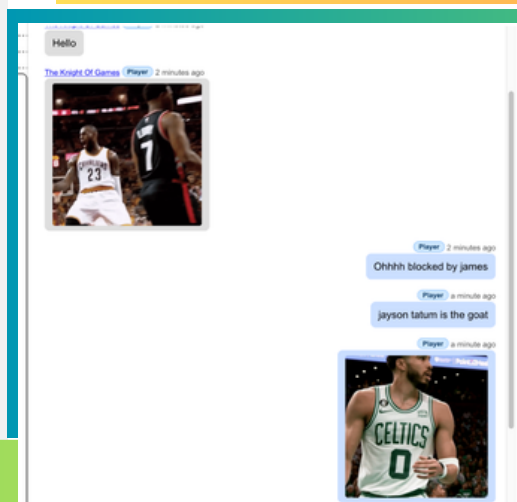
Mafia: Real-time social deduction with roles (Mafia, Cop, Nurse, Civilian), voting phases, and private chat. Design: All games implement a shared GameLogic interface, so Mafia extended this



Leaderboards: Global rankings, win/loss stats, role-aware scoring, daily/weekly/all-time filters, anti-farming scoring. Design: stats are computed and ranked server-side, with built-in anti-farming logic that prevents a malicious user from farming consecutive wins against a victim user



Interactive Chat: @mentions, GIF support via Giphy API, Markdown formatting, private Mafia channels. Design: GIF API keys stay on the server. Mafia-only chat channels are enforced server-side a cheating client can't subscribe itself.



Nim Leaderboard

Nim

Number Guesser

Mafia

Daily Weekly All-Time

. Developed By: Alexander Khazzam, Juhyun Kim, Yuvraj Gohil, Donovan Murray (Group 511)

GameNite!

Group 512: Aryaman Wade, David Yu, Wilson Wei, Olivia Li

GameNite is a real-time multiplayer web platform where users can play games like Nim, Number Guesser, and Push Your Luck, add friends, create friend/invite-only matches, and place wagers with in-game currency. This update also features a complete UI refresh to improve user experiences.

<https://spring26-project-spring26-project-group-yycf.onrender.com/login>

New Features

Friends System

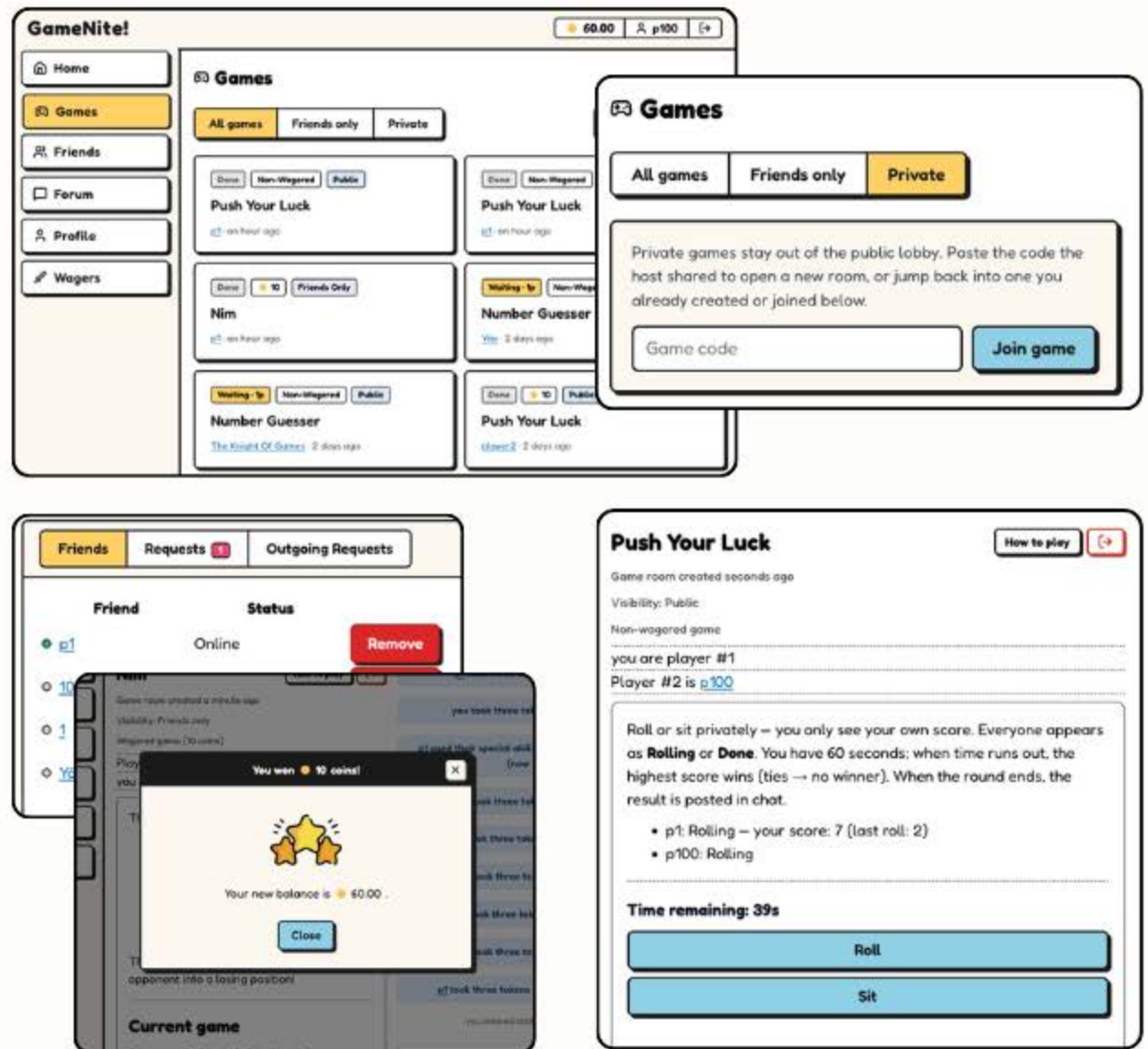
- Send/accept requests, remove friends, view online/offline status
- Create friend and invite-only/private games

Game Updates:

- Nim random special abilities
- Multiple guesses + hints in Number Guesser
- Push Your Luck dice game with timer

Wagering

- wager in-game currency on game results and view wager history



What's next?

The next step is for our team to continue improving overall UX and refining existing gameplay using player feedback. With the MVP in place, we have a strong base to add more complex games and expand the wagering feature with more bet formats and purchasable in-game assets. We'll also focus on scalability and reliability to plan for an expanding user base: reusable architecture patterns, deeper test coverage, and better logging/monitoring to reduce regressions as complexity grows

Tech Stack & Decisions



- extended behavior without tightly coupling systems, making testing, iteration, and future changes safer.
- ensured accurate real-time status updates by marking a user as online when their first socket connects, and only marking them offline when their last socket disconnects.

CodeNames & Friends

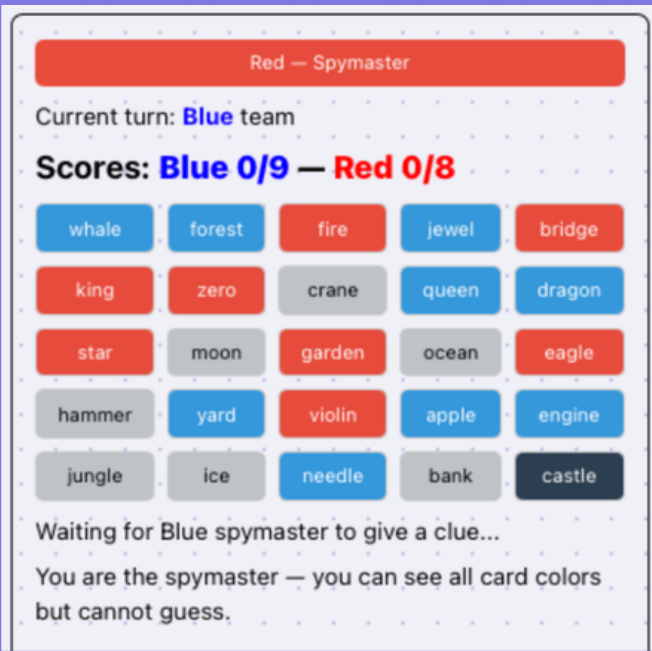
Rikio Suzuki, Renke Cui, Tara Standard, Rachel Chen

Project Overview

GameNite is a social gaming website that allows users to interact through games and messages. This section provides the information needed to build, deploy, and use the application.

New Features

1. CodeNames
2. Leaderboard
3. Friend/Social System



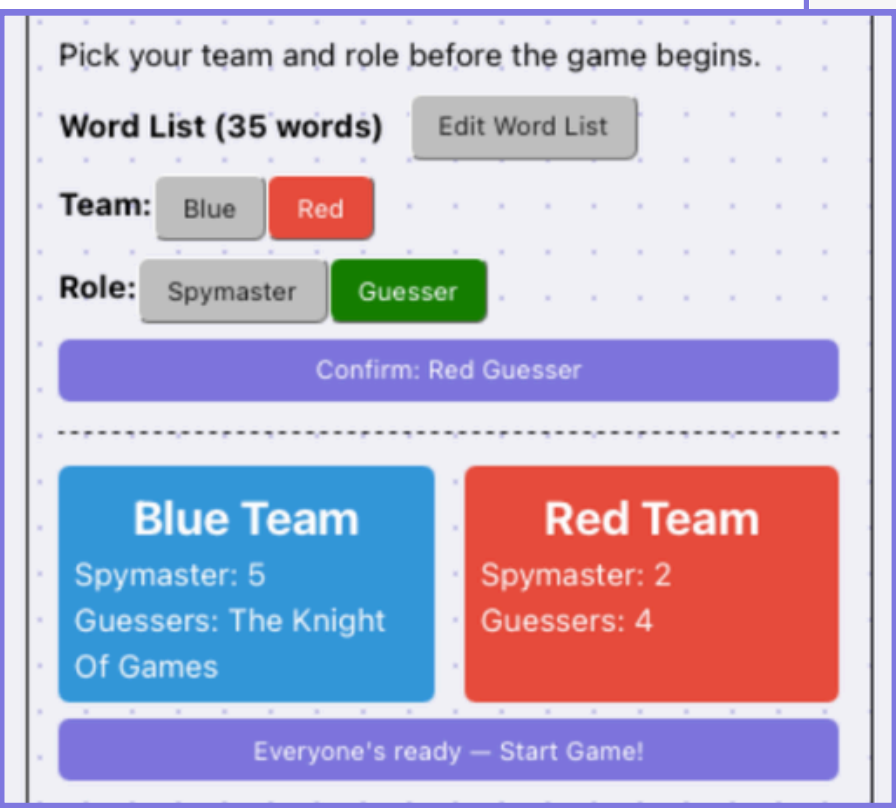
Tech Stack/Design Decisions

Frontend: React and TypeScript
Backend: Node.js and Express,
API endpoints
Database: MongoDB, Supabase

- Decoupled State vs View
- Reusable Game interface
- API calls for social and leaderboard features
- Hybrid Storage Strategy

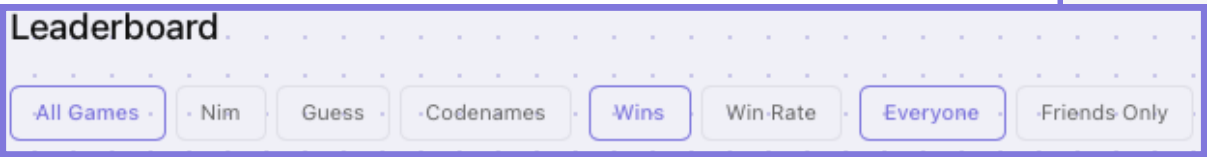
CodeNames

- Multiplayer (4-8 players)
- Team-based roles: Spymaster & Guessers
- Custom word bank support
- Turn-based gameplay with win conditions
- Rematch functionality



LeaderBoard

- Tracks wins, losses, and win rate
- Game-specific performance breakdown
- Filtering: By game, By friends, By win rate or total wins
- Pagination support



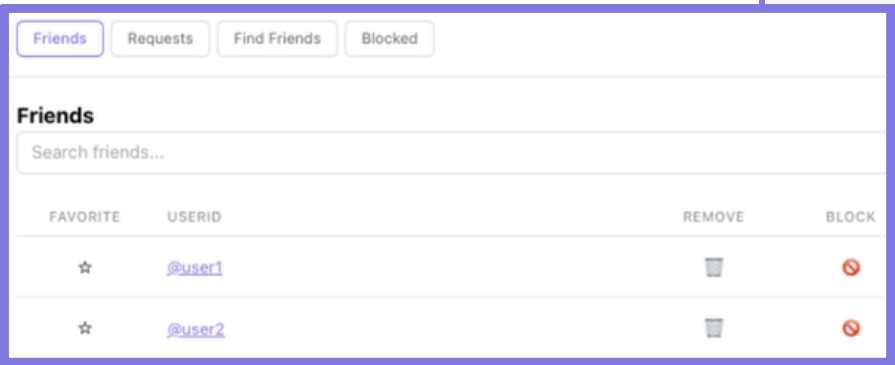
RANK	PLAYER	GAME	WINS	LOSSES	WIN RATE
1	The Knight Of Games	All	4	2	66.7%
2	Yāo (you)	All	2	4	33.3%
3	Frau Drel	All	1	1	50.0%
4	Sénior Dos	All	1	1	50.0%
5	b	All	1	0	100.0%
6	a	All	1	0	100.0%

Why these Changes?

Codenames & Friends is an improved social gaming platform. We expanded the games catalog by adding Codenames as a playable online board game. To increase the competitiveness of the site, we added a leaderboard for players to compete for the better record. Finally, to improve the socializing capabilities, players will now be able to add other users as friends.

Friend/Social System

- Friend requests + management
- Private messaging
- Favoriting friends
- User blocking system



Add Friends

Search for a user by their userID and send them a friend request.

user0	Search
NAME	USERID
The Knight Of Games	@user0
Send Request	

What's Next

- Further improve profile customization capabilities
- Add game lobby creator role to further control privacy
- Add more games to continue expanding the game catalog

Links

GitHub Repository:

<https://github.com/neu-cs4530/spring26-project-spring26-project-group-513.git>

Live Demo:

<https://spring26-project-spring26-project-group-pdzk.onrender.com/login>

Team 514

New Features!

- Friends & DMs
- Private Lobbies
- Leaderboards & Stats

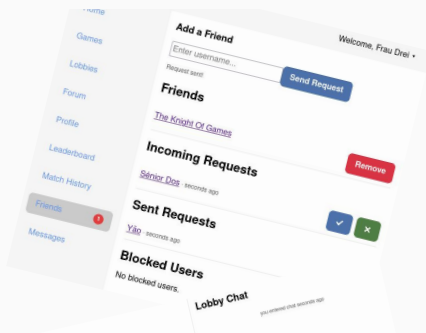
Why these changes?

We thought GameNite lacks the social features needed to foster a sense of community

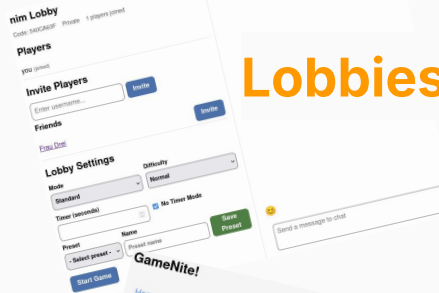
What's Next

- Real-time games
- In-game voice channels
- Team games

Friends & DMs!



Lobbies!



Leaderboard!

Rank	Player	Wins	Losses	Games	Win Rate
1	yihan01	1	0	1	100.0%
2	user0	3	2	5	60.0%
3	user2	2	3	5	40.0%
4	yihan	0	1	1	0.0%



Render

<https://spring26-project-group-514.onrender.com/login>



GitHub

<https://github.com/neu-cs4530/spring26-project-group-514>

Tech Stack & Designs

- **Real-time first:** **Socket.IO** powers live timers, lobby chat, and notifications
- **Storage:** **MongoDB** via Keyv, with in-memory fallback for local dev

Tri Khuu, Logan Chandler, Yihan Luo, Tasmey Makwana



GameNite! Group 515

Driving user retention through persistent progress, customization, and sense of community

Team Members: Niraj Nadarasa, Aditya Anantharaman, Phineas Wormser, Jonathan Cheng

Feature 1: Trophy Rankings

Leaderboards

Track top players by game and see where you rank.

Total trophies Daily activity Nim Number Guesser

Total trophy leaderboard

Showing top 10

Rank	Player	Trophies	Wins	Losses
1	Yao	60	2	0
2	user5	54	3	1
3	you	53	3	5
#4	aditya	52	2	0
#5	S�nior Dos	31	2	1

Competitive progression with persistent trophy tracking

Players earn or lose trophies based on game outcomes, with support for total, per-game, and daily leaderboards. Users can also view trophy totals and win/loss summaries directly on their profiles.

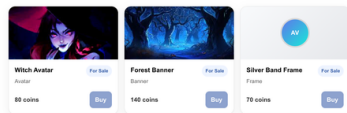
Feature 2: Coins and Cosmetics

Cosmetic Shop

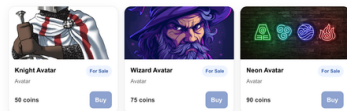
Buy avatars, banners, and frames with coins, then equip them to show on your profile.
Seasonal Cosmetics Time Remaining: 1d 2h remaining

All Avatars Banners Frames

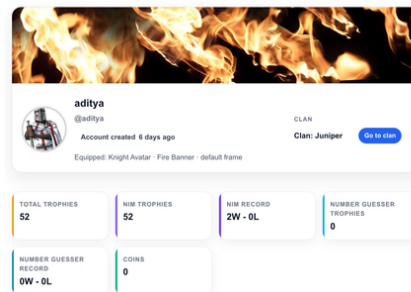
Seasonal Cosmetics



Standard Cosmetics



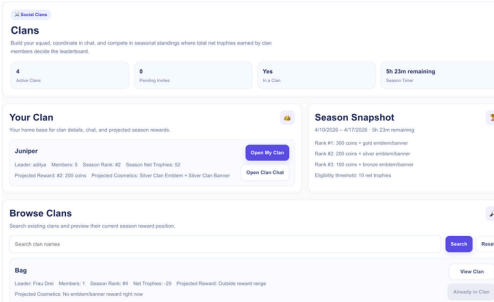
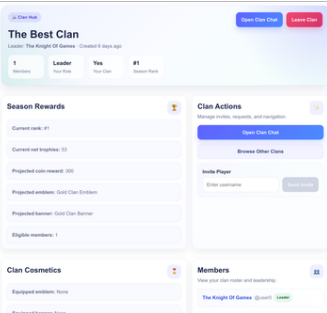
Profile for aditya



A coin economy tied to gameplay and customization

Players can claim coins, create wagers for games, view transaction history, and spend earned currency on avatars, banners, and frames. Purchased cosmetics can be equipped from the shop or profile.

Feature 3: Clans and Seasons



Team-based competition with seasonal rewards

Players can create or join clans, manage roles, chat in real time, and compete on a season leaderboard. Top clans earn rewards based on seasonal net trophy performance and eligible member contribution.



More Info

Tech Stack

Frontend: React, TypeScript

Backend: Node.js, Express

Data / Persistence: MongoDB

Realtime: Socket-based chat & game updates

Testing: Vitest, Playwright

Deployment: Render

Design Decisions

- Server-authoritative progression logic for trophies, coin balances, wagers, and seasonal rewards
- Shared types across client and server to reduce integration mismatches
- Separated UI pages and service layers to keep the frontend focused on interaction and the backend focused on logic

What's Next

Coin Monetization: Allow players to purchase coin bundles with real money to expand the in-game economy

More Games: Add new multiplayer games so that progression and wagering feels fresh

Tournaments: Introduce bracket-style tournaments where players or clans compete for trophies, coins, and exclusive rewards

Try It Out!

Live Demo: <https://spring26-project-group515.onrender.com/login>

Source Repo: <https://github.com/neu-cs4530/spring26-project-group515>

